

Kostiantyn Pysanyi

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a strong research foundation and hands-on experience delivering robust end-to-end ML systems. Specialized in deep learning and computer vision for medical imaging across various modalities.

TECHNICAL SKILLS

ML & CV: PyTorch/Lightning, NumPy, TensorFlow, JAX, Scikit-learn, OpenCV, ITK, pyclesperanto, Detectron2, MMEEngine/MMCV/MMDetection, Albumentations, XGBoost, Pandas, CVAT, Transformers/TIMM/Huggingface
Experimentation & Tracking: MLflow, Tensorboard, Optuna, DVC, Matplotlib/seaborn/Plotly
Infrastructure & Deployment: Linux, Docker, Kubernetes, Prefect, CI/CD (GitLab, Bitbucket), AWS, Triton/TensorRT
Backend & Dev Tools: Flask, REST, FastAPI, OpenAPI, FHIR, Git, SQL, MQTT
Languages: Python (Professional), C++17, TypeScript (Proficient), Java, MATLAB (Basic)

EXPERIENCE

- Machine Learning & Computer Vision Engineer – Medical Imaging** June 2023 – July 2026
RAYLYTIC GmbH Full-time
- Developed **3D instance segmentation and image registration** for light-sheet microscopy images in a liver cancer research project.
 - Built an end-to-end 2D–3D registration pipeline, including mesh processing and alignment, for 3D implant pose estimation from X-rays; improved acceptance rate by 45% and reduced error by 15%; productionized it with **GPU-accelerated Docker** and **Prefect workflow orchestration**
 - Delivered production-grade **segmentation, detection, and classification** models for X-ray imaging; improved accuracy (mAP, Dice, ROC) by 20–50% over baseline and reduced training time 8-fold
 - Implemented an **MLOps** pipeline with **DVC** data versioning, **MLflow** experiment tracking, and automated evaluation
 - Migrated segmentation model serving to a centralized **Triton** Inference Server–based **REST API**, cutting latency by 50%
- Freelance Computer Vision Engineer** Feb 2026 – Mar 2026
Smoothie.de / Sterlix GmbH Contract
- Developed depth-map–based camera obstruction detection for real-time retail anti-theft systems
- Research Engineer** January 2022 – March 2023
Data Science Research Group Part-time
- Developed ML model for bio-signal classification for pain assessment in OP scenarios without patient feedback
 - Designed data pipelines and training workflows for a ticket-routing classifier, transforming unstructured text into structured features and streamlining dataset updates and model evaluation for industry partners
- Internships**
- ML Thesis Student at RAYLYTIC GmbH – (Oct 2022 – Feb 2023): Developed anatomical landmark localization model
 - ML Intern at Prudsys GmbH – (Dec 2021 – Feb 2022): Automated analytic workflows and data preprocessing pipelines
 - Autonomous Driving Intern at IAV GmbH – (Sep 2021 – Nov 2021): Investigated inverse reinforcement learning for self-driving
 - Student Research Assistant at WHZ – (Jan 2020 – Dec 2021): Developed **semantic segmentation for automated tissue-density analysis of 3D lung CT scans**; built an autoML library and modeled COVID ICU admission

EDUCATION

- B.Sc. Data Science with distinction (1.1 / 1.0)** Oct 2019 – Apr 2023
University of Applied Sciences Zwickau Zwickau, Germany
- M.Sc. Civil Engineering (wGPA: 90.12/100)** Sep 2014 – Jun 2020
Odesa State Academy of Civil Engineering and Architecture Odesa, Ukraine

PUBLICATIONS & AWARDS

- EFORT 2026:** Validation of a 2D-3D Image Registration Method for TKA Implant Pose Estimation. Pysanyi et al.
DWG 2023: Automatic determination of lumbar segmental lordosis based on an AI algorithm. Pysanyi et al.
1st Place: National Strength of Materials Olympiad (Ukraine, 2018)

LANGUAGES

German (C1, TestDaF), English (C2, EF SET Certificate)